

Blind over-the-air test of whether the phase noise/jitter performance of John Ackermann N8UR's MVR low cost GPSDO is good enough for FST4W

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As well as this summary document there is an Excel spreadsheet with the data and calculations.

Method

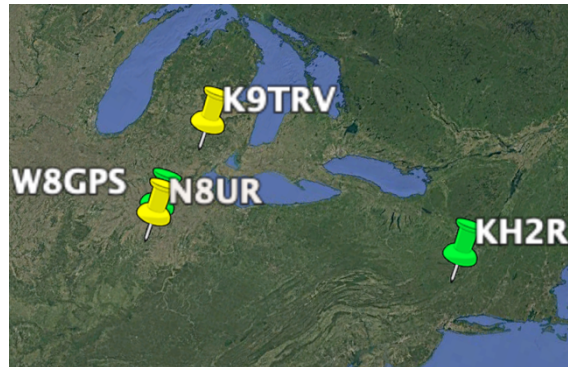
N8UR and K9TRV each have a pair of identical RX888 MkII receivers and are decoding FST4W from two WSPRSondes that have GPSDOs (Bodnar mini at W8GPS and unknown type at KH2R).

1. N8UR using W8GPS

km N of N8UR, consequently the test was of wave signal a) when the Doppler shift was visual check of the spectral spread on the WSPRSonde [dashboard](#) showed surface dominant propagation mode because of low (16:24 to 18:30 UTC 8 September 2026

2. N8UR and K9TRV using KH2R

km bearing 258° from KH2R and K9TRV is a ~800 km on bearing 278°. These paths are, beyond surface wave, and its minimal spread. Ionospheric reflection via F2 layer much Doppler shift and frequency spread to be useful as comparison medium. However, reflection via the (normal) E region introduces very low Doppler shift and frequency spread. Fortunately daytime 30 m propagation from KH2R to N8UR and K9TRV was, at the time, via the E region (13:36 to 20:52 UTC N=70).



W8GPS is ~32 80 m surface zero and b) Grafana wave to be the spectral spread N=64). N8UR is ~866 similar distance of course, well frequency introduces too

Recognising that the distribution of frequency spread on these paths will not be Gaussian I calculate the **median** and the **median of the absolute deviation** (mad) instead of mean and standard deviation for when Doppler shift was zero.

Results

1. N8UR using W8GPS

KA9Q_0	median 9 mHz	mad 1 mHz	N=64
KA9Q_1	median 9 mHz	mad 1 mHz	N=64

2. N8UR using KH2R

KA9Q_0	median 11 mHz	mad 3 mHz	N=53
KA9Q_1	median 11 mHz	mad 3 mHz	N=55

3. K9TRV using KH2R

K9TRV	median 17 mHz	mad 6.5 mHz	N=30
KA9TRV-4	median 16.5 mHz	mad 7.5 mHz	N=49

Conclusions

At **N8UR**, on both paths surface wave and E region, there is **no difference** in median or median absolute deviation between the frequency spread measurements from the two RX888s with different GPSDOs.

At **K9TRV**, in all honesty, there is **no difference** in median or median absolute deviation between the frequency spread measurements from the two RX888s with different GPSDOs. What is a bit odd is the difference in number of spots meeting the Doppler=0 Hz criterion.

I do not know why the median and median absolute deviations at **K9TRV** are higher than at **N8UR**, whether the longer path length plays a part, or whether different local electrical connections or some other cause. Note SNRs were higher at K9TRV (median ~ -10 dB) than at N8UR (median -27 dB). Always more loose ends that one could follow up...